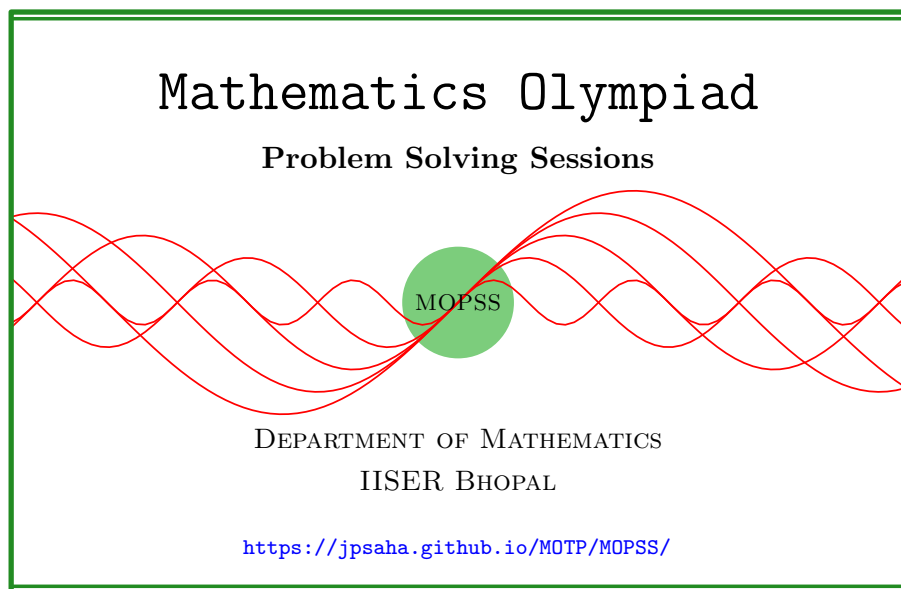


Gauss's lemma

MOPSS



Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from OTIS Excerpts [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for high schoolers in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

List of problems and examples

1.1 Exercise (ELMO 2009, AoPS, proposed by Evan O’Dorney)

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§1 Gauss’s lemma

Exercise 1.1 (ELMO 2009, AoPS, proposed by Evan O’Dorney). Let a, b, c be positive integers such that $a^2 - bc$ is a square. Prove that $2a + b + c$ is not prime.

Walkthrough —

- (a) Consider the quadratic polynomial $p(x) = bx^2 + 2ax + c$.
- (b) Show that its discriminant is a perfect square.
- (c) Use Gauss’s lemma to show that $p(x)$ can be factored into linear polynomials with integer coefficients.
- (d) Note that the roots of $p(x)$ are negative rationals.
- (e) Conclude that $p(x)$ can be factored into linear polynomials with positive integer coefficients.
- (f) Conclude that $p(1) = 2a + b + c$ is not a prime.

Solution 1. Consider the quadratic polynomial $p(x) = bx^2 + 2ax + c$ with integer coefficients. Since its discriminant is a perfect square, it follows that its roots are rational, that is, it can be factored over the rationals. By Gauss’s lemma, $p(x)$ can be factored into linear polynomials with integer coefficients. Since the leading coefficient of $p(x)$ is positive, it follows that it can be factored into linear polynomials with integer coefficients and having positive leading coefficients. Note that the roots of $p(x)$ are negative rationals. This proves that $p(x)$ can be factored into linear polynomials with positive integer coefficients. Noting that $p(1) = 2a + b + c$, it follows that $2a + b + c$ is not a prime. ■

Remark. Note that in the above, one may prove that $p(x)$ can be factored into linear polynomials with integer coefficients without using Gauss’s lemma, possibly by establishing the lemma in this specific case. In fact, the above problem could serve as an introduction to Gauss’s lemma.

References

- [Che25] EVAN CHEN. *The OTIS Excerpts*. Available at <https://web.evanchen.cc/excerpts.html>. 2025, pp. vi+289 (cited p. 1)